Automation Risk, Redistribution, and Economic Vulnerability: Evidence from the European Social Survey, 2002–2022

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Abstract

Does occupational exposure to automation risk increase demand for government redistribution? Thewissen and Rueda (2019) found that workers in occupations more exposed to routine task automation support redistribution more, on European Social Survey data through 2012 across seventeen countries. We derive the hypotheses from an insurance model solved in closed form. In the model, displacement risk raises demand for redistribution at every level of risk aversion. We rebuild the design from raw records across all eleven waves, more than 260,000 respondents in 32 countries, with two independently constructed exposure indices. We find that their result replicates through 2012 and disappears in the aggregate from 2016 onward, and that exposure predicts financial strain in the same waves in which it predicts redistribution support. Across countries, though not within them, that link is weaker where flat rate transfers are generous and stronger where employment protection and net inequality are high. For redistribution itself, benefits that track prior earnings amplify the gradient between countries, while generous flat transfers and net inequality flatten it within countries. In the most recent wave, exposed workers report more financial strain in eight countries but demand more redistribution in only two. Hence, the political consequences of technological displacement depend on how domestic institutions distribute the economic strain it brings.

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1 Introduction

Recently, interest has renewed in the role of economic self-interest in explaining citizens' political attitudes, preferences, and behavior. Developments in digital and physical systems that operate using artificial intelligence have drawn attention to the political effects of job automation and its effect on workers' displacement. Extensive literature shows that an occupation's measured automation risk correlates with several political attitudes and behaviors. For example, higher routine task intensity is associated with greater support for redistribution, particularly among higher income workers (Thewissen and Rueda, 2019). Higher occupational automation risk is also associated with radical right voting among workers who report coping with their current income (Im et al., 2019). In contrast, in Spain, workers concerned about technological change favor slowing its adoption, while neither measured exposure nor subjective concern strongly predicts redistribution preferences (Gallego et al., 2022a). The benchmark finding has prompted further work (Im et al., 2019; Kaihovaara and Im, 2020; Gallego et al., 2022b; Heinrich and Witko, 2022).

Occupational automation risk measures how readily technology can perform the tasks in an occupation and thereby threaten employment and wages. In traditional models of political economy, people demand public insurance against those losses. Scholars of automation and redistribution have tested this prediction as a direct link from automation risk exposure to policy preferences, but have not examined how institutions buffer or amplify the material risks of lower wages and longer unemployment. Economic insecurity can raise demand for protection (Margalit, 2013, 2019). Yet existing accounts disagree or stay silent on how institutions strengthen or weaken the link between risk and redistribution (Moene and Wallerstein, 2001; Iversen and Soskice, 2001; Korpi and Palme, 1998). We develop a theoretical model of social insurance to derive predictions about how the risk of losing a job affects support for redistribution under different benefit rules. We derive an increase in support as job loss risk rises. With flat payments, this increase is smaller than with benefits based on past wages. We incorporate employment protection through displacement risk and net inequality through relative income.

Using two exposure indices, we analyze all eleven waves of the European Social Survey from

2002 to 2022, covering more than 260,000 respondents in 32 countries. We rebuild the benchmark design from raw ESS records harmonized to a common format and extend coverage a decade beyond the published window. Since institutions differ across countries and change over decades, we use both kinds of variation in our test. We estimate slopes for each country and for each country and year from a single multilevel model. Our reference measure is the automation probability of each occupation from Frey and Osborne (2017). For robustness, we also use the routine task intensity index of Goos et al. (2014), the closest analog to the canonical benchmark of Thewissen and Rueda.

Whichever countries we include, we no longer find a positive aggregate relationship between automation exposure and redistribution support from 2016 onward. Changing country coverage or sample composition does not explain the decline. Countries differ far more in this association than it varies over time. We can recover the positive association found in earlier studies through 2012. Refitting the published estimator of Thewissen and Rueda (2019) on their replication data gives us almost exactly their baseline, and we recover the positive association in the harmonized panel over 2002 to 2012. We estimate the decline directly rather than infer it from subsamples. Its magnitude after 2016 exceeds that of the pooled historical gradient in absolute terms. The benchmark countries saw measured routine exposure fall over this period, and some attenuation may stem from occupational measurement error after the ISCO-08 revision (van Hoorn, 2022).

With the main measure of automation risk, the gap in support for redistribution between workers facing high and low risk narrows as flat benefits or net inequality rise within a country. The gap is wider in countries where unemployment benefits depend more on prior earnings. Workers in safer jobs also support a higher redistributive tax in the model because it expands flat benefits. Thus, their gap with exposed workers narrows under flat benefits and stays wider under benefits tied to past wages and independent of that tax choice. Automation exposure predicts financial hardship in the same waves in which it predicts redistribution support. Note that the strain gradient in the full sample is significant in six of the seven waves where the redistribution gradient is, and in none where it is not. Exposure also predicts expected job loss and perceived job insecurity in the rounds that include them. Net of the business cycle, institutions moderate the link between exposure and strain between countries rather than within them.

Related work has documented ESS associations between automatability and redistribution using other exposure measures and designs (van Hoorn, 2022). Studies of specific policy areas document responses to technological risk (Gallego et al., 2022a; Busemeyer and Tober, 2023). The literature also covers welfare state attitudes under automation risk (Busemeyer and Sahm, 2022). Other work documents the electoral response to state automation policy (Gingrich, 2019). Busemeyer and Tober (2023) report institutional moderation of the opposite sign to the one predicted here, and Jeffrey (2020) finds redistribution preferences flat or falling under randomised automation vulnerability. The first of our three contributions is a theoretical correction to the benchmark literature's informal assumption of a positive interaction between income and risk in redistribution demand. Our formal model shows that this interaction contradicts standard estimates of risk aversion and formally rules it out. The interaction requires a restriction on parameters rather than following naturally from insurance logic.

Second, we examine how welfare state design affects political coalitions. Since Korpi and Palme (1998), political economists have often treated flat rate transfers and earnings related benefits as varying points on a single spectrum of welfare generosity. In our model, the politics of risk in a country depends on benefit structure because the designs move the gradient in opposite directions. Flat transfers provide a universal floor that buffers economic anxiety and flattens political demand for redistribution. In contrast, professionals with high earnings still have much more to lose from automation under benefits that scale with a worker's prior earnings and preserve income hierarchies.

How general was the benchmark association? Earlier work estimates one such coefficient, and the moderation literature explains that composite. We make our third contribution by decomposing it into slopes for each country and year. We find a strong association in some countries, an association that fades in most during the 2010s, and no association at all in others. The benchmark association was never general. Our finding rests on the exposure indices meaning the same thing across countries and waves, and the attenuation results above put that assumption under strain.

The rest of the paper is organized as follows. First, we develop a formal insurance model that informs the hypotheses of the study, with the derivations in Appendix 8.1. Next, the research

design of the study is discussed, namely the panel, the exposure measures, and the estimator. Then, the paper proceeds to describe the results of the benchmark replication, the attenuation after 2016, and the slopes by country and year. After that, the mechanism analyses trace exposure to perceived vulnerability and test the four institutional predictions. Finally, the paper ends with a discussion and a conclusion, and the appendix reports the results for the routine task intensity index and further robustness checks.

2 Theory

2.1 Comparative Political Economy of Redistribution

We begin with two accounts of welfare state development and redistribution. Wilensky's modernization account explains welfare state expansion as a response to industrialization, economic growth, and demographic change (Wilensky, 1974). Instead, Stephens' power resources account explains welfare state development as sustained by organized labor and its political arm, including union strength, socialist parties, and collective bargaining (Stephens, 1979).

Moene and Wallerstein (2001) show that preferences over social insurance and redistribution depend on the distribution of risk and income, not on inequality alone. Iversen and Soskice, in their asset theory, treat preferences over social policy as a function of exposure to labor market risk and the portability of workers' skills (Iversen and Soskice, 2001). Korpi and Palme have stressed an institutional feedback in their paradox of redistribution (Korpi and Palme, 1998). In their account, encompassing welfare states can reduce poverty and inequality because broad social programs build larger political coalitions for redistribution.

2.2 Job Automation and Economic Threat

Automation can alter perceptions of economic threat, which in turn can change political attitudes. van Hoorn (2022) documents a positive automatability–redistribution association in the ESS using a risk measure defined at the occupation level. Heinrich and Witko (2022) connect technology threat to candidate preferences in the United States (see also Im et al., 2019; Kaihovaara and Im, 2020; Gallego et al., 2022b). Additionally, research on economic shocks links

perceived insecurity, especially anticipated unemployment, to changes in political and social preferences (Margalit, 2019, 2013). The related literature on economic vulnerability and political psychology treats perceived threats as a potential source of support for protective policies (Hacker et al., 2013; Rogers, 1975).

2.3 Automation Risk as Conditional Redistribution Demand

The rationale for the benchmark prediction is that occupational exposure can signal future income and employment losses, so that respondents in routine, automatable occupations may anticipate displacement or downward mobility and consequently find redistribution and social insurance more attractive. In other words, routine task automation risk increases demand for redistribution in the benchmark account (Thewissen and Rueda, 2019). Hence, we expect automation exposure to be positively associated, on average, with support for redistribution.

As economies develop and employment structures and institutions change, people understand occupational risk differently. Workers can feel more or less threatened by the same exposure during recessions, recoveries, and inflation. People and firms also adapt as automation diffuses and change their expectations as tasks shift across occupations. To the extent that technological change advances beyond the computerization of routine work, routine task measures may lose political relevance. Thus, we expect the automation–redistribution association to vary across survey waves, without assuming either a constant association or a monotonic movement in one direction.

One essential question is why the same exposure may have different consequences across countries. Welfare states and labor market institutions affect whether workers bear automation risk privately or have it collectively insured. Where workers are strongly organized and bargaining is broad, they have power before market risk becomes individualized. Moreover, generous unemployment protection, social spending, and redistribution can buffer the material and psychological consequences of exposure. Hence, exposure may generate greater demand for redistribution where protection is weaker or inequality is higher, because workers face a risk that is less insured.

We use perceived vulnerability to trace how exposure may lead to redistribution demand.

The idea is that exposure may first appear in expectations of job loss, financial strain, and insufficient income. Drawing on these theoretical expectations, we advance three hypotheses, with H_1 predicting more support for redistribution as occupational automation risk increases, H_2 predicting variation in this association across time, and H_3 predicting variation across countries. Beyond these hypotheses, M_1 asks whether perceived economic vulnerability partly links automation risk to redistribution demand. In M_2, we examine whether welfare state and labor market institutions moderate the association between exposure, vulnerability, and redistribution.

2.4 An Insurance Model of Redistribution Demand

In this section, we develop a model of a worker who values a program of taxes and transfers according to her employment prospects. She values the program for what it pays her now if she expects to keep her job, and for what it would pay her then if she expects to lose it. In what follows, we make the second valuation explicit enough to sign the channel through which automation exposure acts, with the algebra in Appendix 8.1.

Let us write her income relative to the mean as ξ, and let p be her displacement probability. Displacement is a temporary spell, during which her income depends on four quantities that welfare states set separately. Let μ be the expected share of the period out of work, and ρ her reentry wage as a fraction of her old wage. Additionally, let b be the benefit rate tied to earnings and B the benefit duration. We can express her retention share δ as a function of these four quantities,

$$\delta \;=\; (1-\mu)\,\rho \;+\; b \cdot \min(\mu, B). \tag{1}$$

After the spell, she receives wage income scarred by the interruption, which is captured by the first term. The second term represents her benefit income while the entitlement runs. Since a different institution produces each of these four components, an index cannot identify which component produced the result.

Also, assume that transfers are financed by a linear tax τ and that she receives θ_g in work and θ_b out of work. We will think of δ as collecting the payments proportional to her own past earnings, because she receives them whatever tax rate she votes for. Payments proportional to

the tax rate belong to θ_b and pay her the same amount as everyone else. Hence, unemployment insurance tied to earnings and social assistance fall on different sides of this distinction.

The worker chooses her preferred tax rate by weighing the two states by their probabilities and her distaste for the gap between them. Her risk aversion σ measures that distaste. Maximising her expected utility, we obtain the following closed form,

$$\tau^* = \frac{\xi\,(1-\kappa\delta)}{(\xi-\theta_g)+\kappa\,(\theta_b-\delta\xi)}, \qquad \kappa \equiv \left[\frac{(1-p)\,(\xi-\theta_g)}{p\,(\theta_b-\delta\xi)}\right]^{1/\sigma}, \tag{2}$$

where κ is the ratio of her consumption in one state to her consumption in the other. We interpret the preferred tax rate as her demand for redistribution, the disposition measured by the ESS item asking whether government should reduce differences in income levels. In other words, a rise in τ^* represents greater support for redistribution. Define the exposure–redistribution gradient as the change in τ^* per standard deviation of automation exposure.

Note that the retention share influences her demand through the consumption ratio. This means that a higher δ leaves her more income after displacement and brings the two states closer together.

> *Risk raises demand unconditionally.* A worker's preferred tax rate rises with her displacement probability, for every level of risk aversion and every setting of the remaining parameters (Proposition 2).

Intuitively, a higher chance of displacement shifts part of her expected life from the state where she funds the transfers to the state where they pay her. Applied to occupational exposure in H_1, the assumption is that, with the wage and severity channels fixed, exposure raises her perceived chance of displacement.

> *Benefit earnings relatedness amplifies the gradient.* Exposure raises her preferred tax rate more where a larger share of a displaced worker's former earnings is replaced (Proposition 8).

> *Flat generosity buffers the gradient.* Exposure raises her preferred tax rate less where transfers pay the same sum regardless of former earnings (Proposition 8).

The distinction between these two predictions depends on one condition, namely whether the income a worker keeps after displacement is proportional to prior earnings. First, consider a flat transfer. This matters most to the worker with least at stake and compresses the gap on which exposure acts. In contrast, if her cover tracks her earnings, her loss stays proportional to

her old wage and a worker with more to lose continues to have more to lose as exposure rises. The same proposition signs both predictions in Table 1 because an identity in the model ties the two derivatives together. Hence, a country cannot come out on the wrong side of both predictions, and the pair is a single test.

> *Employment protection steepens the gradient, but only above log utility.* Exposure raises her preferred tax rate more where the baseline chance of displacement is lower, provided risk aversion exceeds one. At log utility the effect is exactly zero (Propositions 3 and 4).

Let us discuss how this prediction relates to the empirical setting. While the result concerns curvature, that is, how the exposure slope itself changes as baseline risk changes, we do not estimate curvature directly in the panel. We use a linear interaction across levels to ask whether slopes differ between countries with stricter and looser protection. The interaction tests only the sign implied by the curvature result, given that employment protection shifts baseline risk. A worker who already expects to lose her job has little left to learn from a further rise in risk, while a worker who expects to keep it revises sharply. Note that every country in this panel is below the displacement probability of one half at which this reasoning holds.

> *Net inequality flattens the gradient.* Exposure raises her preferred tax rate less where relative income is more dispersed (Propositions 5 and 6).

The model predicts an ambiguous income effect in general, since risk aversion determines the balance between two opposing forces. First, at high risk aversion, insurance behaves as a normal good and richer workers buy more of it, as in the region described by Moene and Wallerstein (2001). On the other hand, at low risk aversion, richer workers want less because the redistributive motive dominates, following the benchmark of Meltzer and Richard (1981). Meta analyses put relative risk aversion near one, placing this panel in the second region. Hence, rising inequality crowds out the insurance motive, and we predict a flatter gradient.

For exposure to appear in her demands on government, it ought first to appear in her account of her own position, since risk in this model is a perception. We test this link in M_1 and the four institutional results above in M_2.

2.5 Measuring Routine Task Automation Risk

Computers and machines can more readily perform codifiable, repetitive tasks, which is why this literature rates occupations by the routineness of their task content (Autor and Salomons, 2018). We use the automation probability of Frey and Osborne (2017), built from engineering bottlenecks. We also use the routine task intensity (RTI) index of Goos et al. (2014). The two measures correlate at 0.598 in our sample. Both assign their highest values to clerical, production, and less skilled service occupations, where workers also tend to have lower pay and less employment security.

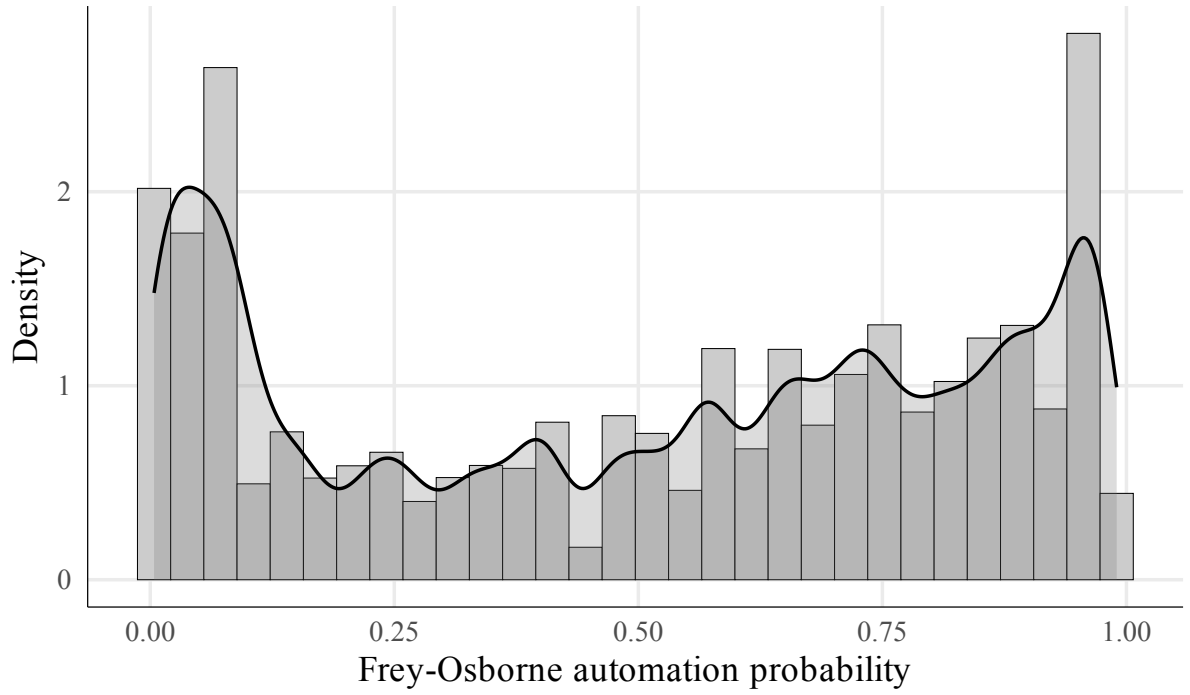
This is the construct used in the redistribution literature, and we retain both operationalizations throughout the paper. The idea is that, if perceived vulnerability tracks labor market precarity, we should observe greater vulnerability as routine task exposure rises among respondents in the occupations identified as exposed. Hence, we expect the automation–redistribution association to be strongest where institutions protect least against that vulnerability.

3 Research Design

3.1 Data and Measures

To assess the extent of the association between automation exposure and support for redistribution, we use a harmonized panel of eleven European Social Survey (ESS) waves from 2002 to 2022. The panel includes 32 countries and roughly 260,000 respondents before listwise deletion. The dependent variable of the analysis is the standard ESS item asking whether government should reduce income differences. The item is scored from one to five, and higher values mean greater support for redistribution. We follow Thewissen and Rueda (2019) and limit the main analysis to employed and unemployed members of the active labor force.

Figure 1: Distribution of Job Automation Risks in the ESS Sample



In a graphical analysis of the Frey–Osborne automation probabilities for ESS occupations (Figure 1), the density of the measure resembles the corresponding U.S. distribution in Frey and Osborne (2017), consistent with a successful occupation match.[1] Additionally, the figure reveals that the observations cluster toward the ends of the 0–1 scale, because the measure is a probability.

Following Hout (2004), we convert responses on the income bands of ESS show cards to monetary midpoints and, for the open top band, estimate a Pareto mean from the observed counts in the top two categories.

Since rounds 1–5 use ISCO-88 and rounds 6–11 use ISCO-08, we harmonize both exposure measures across the two occupation systems. We validated the harmonization by joining the raw panel to the original replication data of Thewissen and Rueda (2019) at the respondent level for rounds 1–6. We recover RTI exactly, Frey–Osborne probabilities almost exactly, and the redistribution outcome and core demographic variables almost perfectly.

Regarding economic vulnerability, financial strain is available in every wave, and we code it as whether the household finds it difficult or very difficult to cope on its current income. Among

[1] The match covers almost 90% of potential cases, and matched and unmatched cases do not differ systematically.

labor market participants, we also use expected job loss (a binary indicator for expecting to lose one's job in the next twelve months) and expected income insufficiency (a binary indicator for expecting income too low to meet needs) from the 2008 and 2016 welfare attitudes modules, and perceived job insecurity (a scaled measure of the perceived likelihood of job loss) from the 2004 and 2010 work modules.

At the country level, we examine 11 institutional moderators (see the institutional measures dictionary in the appendix). Depending on availability, the moderation models contain 25–32 national clusters. Six annual series come from the Comparative Political Data Set (CPDS; Armingeon et al., 2025), and four additional series come from their source APIs. The eleventh moderator, benefit earnings relatedness, measures how far unemployment benefits depend on prior earnings. We compute it as the difference between the OECD net replacement rate of a claimant at the average wage and that of a claimant at 67% of it, at month six of a spell. The net replacement rate series cannot separate flat rate systems from systems tied to earnings because it is evaluated at a single earnings level. Fetscher (2023) measures that structure from replacement rates across earnings levels, deriving an indicator of benefit concentration from the same OECD calculator over ranked income groups, in the tradition of Beramendi and Rehm (2016) and the progressivity index of Kakwani (1977). Ours is a deliberately coarse version that we build from two points such that the series runs annually for every country and year in the panel. However, in systems where a statutory ceiling is the binding feature, a window closing at the average wage understates earnings relatedness, and ceilings bind above the average wage in many systems.

3.2 Empirical Strategy

For redistribution preferences, we estimate one multilevel model with random effects for each exposure index. The model regresses individual support for redistribution on occupational automation exposure, standardized on the pooled estimation sample, and on the benchmark controls. Each country and year receives its own intercept and its own slope on automation risk, and the slope has global, country, year, and country by year components (equations 3–4).

Our primary models include the baseline control vector of Thewissen and Rueda (2019), for consistency with the published benchmark. Specifically, the baseline vector includes relative

equivalized income, years of education, gender, and age. Current or former union membership, religiosity, and individual unemployment status are also included. In the style of the benchmark study, the appendix tests the gradient against rival explanations at the individual level, one at a time.

We decompose the association between occupational exposure and each outcome of perceived vulnerability into paths through relative income, occupational unemployment risk, and contract precarity. We use a cluster bootstrap of 2,000 replications that resamples countries with replacement. Exposure is not assigned, and the channels are themselves products of occupational sorting. Hence, we make no assumption of sequential ignorability. While the decomposed paths identify no causal mediation, they describe how the association is distributed across measured channels at the individual level, in a single cross section.

Let y_{ict} be the outcome of individual i in country c and survey year t, with AR_{ict} the index of occupational automation risk. Depending on the analysis, y_{ict} is support for redistribution, the primary outcome, or one of the vulnerability indicators described above. We estimate the same specification in all models:

$$y_{ict} = \alpha_{ct} + \beta_{ct} AR_{ict} + \mathbf{X}_{ict}\boldsymbol{\phi} + \varepsilon_{ict}, \tag{3}$$

where α_{ct} is an intercept specific to each country and year, $\mathbf{X}_{ict}$ is the vector of controls at the individual level with coefficient vector $\boldsymbol{\phi}$, and β_{ct} is the slope for automation risk specific to each country and year. It will be useful to write the slope β_{ct}:

$$\beta_{ct} = \gamma_{10} + u_{1c} + v_{1t} + \omega_{ct}, \tag{4}$$

where γ_{10} represents the global baseline slope, u_{1c} and v_{1t} the country and year deviations, and ω_{ct} their interaction. Finally, ε_{ict} is the error at the individual level. We estimate the model by maximum likelihood. Since the country by year components of its random part absorb dependence within each country and year, we apply no clustering correction after estimation, and all reported standard errors are model based. In the figures, the 95% intervals around fitted slope components combine the sampling variance of the fixed slope with the conditional variances

of the relevant country, year, and country by year components. We ignore posterior covariances across components, the standard approximation for intervals around best linear unbiased predictions.

In the models where a perceived vulnerability indicator is the dependent variable, we use linear probability models, one for each indicator. Also, in these models, we use design weights and random intercepts. We estimate the institutional moderation models in the within–between form developed for repeated cross sections of respondents nested in countries observed over multiple waves (Bell and Jones, 2015; Fairbrother, 2014; Bell et al., 2019). Giesselmann and Schmidt-Catran (2019) extended this form to interactions across levels.

Let us write the institutional moderator, standardized and lagged by one year, as $Z_{c,t-1}$, with $\bar{Z}_c$ its country mean. Thus, we can express the institutional moderation models as

$$y_{ict} = \alpha_{ct} + \beta_c\, AR_{ict} + \gamma_B\big(AR_{ict} \times \bar{Z}_c\big) + \gamma_W\big(AR_{ict} \times (Z_{c,t-1} - \bar{Z}_c)\big) + \mathbf{X}_{ict}\boldsymbol{\phi} + \lambda_t + \varepsilon_{ict}, \quad (5)$$

where γ_B gives the moderation between countries and γ_W the moderation within them. Note that the deviation $(Z_{c,t-1} - \bar{Z}_c)$ represents the variation within countries. We include main effects of both moderator components, exposure interactions with contemporaneous unemployment and GDP growth at constant prices, and year fixed effects λ_t. Exposure slopes that vary by country and a random intercept for each country and year are also included. The idea is that, if we left out the random exposure slope, inference on interactions across levels would be overconfident (Heisig and Schaeffer, 2019). By giving each country and year its own intercept, we absorb macro shocks at the wave level that would otherwise understate the standard errors of interactions across levels in pooled cross sections (Schmidt-Catran and Fairbrother, 2016).

We treat four of the moderators as focal because a stated proposition signs each of them through one primitive of the model (Table 1). We combine standardized social spending and realized redistribution in one mean, the flat generosity bundle. In all of our models, the replacement rate, left cabinet share, industry employment share, and trade openness are the same uninteracted controls for each country and year. Naturally, the focal design excludes union density and bargaining coverage, which the model leaves unsigned, and their interactions appear only in the appendix sweep. Since each proposition is a partial derivative that holds the other prim-

itives fixed, we enter the four focal moderators in the primary redistribution model jointly (the appendix retains the versions with one moderator at a time as robustness). Regarding financial strain, its moderation models carry no signed predictions and therefore estimate one moderator at a time, as mechanism diagnostics. For each outcome, the sixteen focal coefficients form the family within which we apply the Benjamini–Hochberg adjustment.

Table 1: Focal moderators and the comparative statics they test

Moderator	Primitive	Result	Predicted sign
Benefit earnings relatedness	δ (retention)	Prop. 8	$+$ amplifies
Flat generosity bundle	θ_b (flat transfer)	Prop. 8	$-$ buffers
Employment protection	p (displacement risk)	Props. 3–4	$+$ steepens if $\sigma > 1$; nil at $\sigma = 1$
Net inequality	ξ (relative income)	Props. 5–6	$-$ flattens
Controls: left cabinet share, industry employment, trade openness, replacement rate			

Given that the ESS samples new respondents in each wave, we decompose each moderator over the country panel (Fairbrother, 2014). Additionally, the term γ_B in the fixed part controls for heterogeneity between countries in the exposure slope that each country receives in the random part. While a random slope alone does not provide this control, Giesselmann and Schmidt-Catran (2019) show that it is required for γ_W to be a genuine within estimator.

In this paper, a coefficient is significant only where its Benjamini–Hochberg adjusted q value meets the $q < 0.10$ threshold within the family of tests its design defines.[2] A reader who prefers a different rule will find the family sizes where each family is introduced, and unadjusted standard errors throughout. Although familywise corrections such as Holm's (Holm, 1979) control the probability of even one false positive, they pay for that guarantee in power. Hence, following Benjamini and Hochberg (1995), we use their procedure to control the false discovery rate, the expected share of false positives among the coefficients a family declares significant. To preserve the registration of the confirmatory vulnerability table (Table 6), we keep its stricter Holm adjustment within each outcome–wave pair.

[2] Within a family of m tests, the adjustment ranks the raw p values $p_{(1)} \leq \cdots \leq p_{(m)}$ and rejects the k hypotheses ranked lowest at level q, where k is the largest index with $p_{(k)} \leq (k/m)\, q$. Thus, the q we report for a coefficient is its adjusted p, the smallest level at which the adjustment would reject it, and it is never below the raw p.

4 Attenuation and Heterogeneity across Countries in the Automation–Redistribution Association

Before any harmonization of ours enters, we refit the published estimator of Thewissen and Rueda (2019) on their own replication data, with their sample restrictions and a country random intercept. We obtain a Frey–Osborne coefficient of 0.162 versus the published 0.162 ($N = 59,997$) and an RTI coefficient of 0.042 ($N = 64,639$). In our harmonized ESS panel, we recover the Frey–Osborne association over the original 2002–2012 window after adopting the benchmark outcome, restriction to the labor force, relative income control, and exposure indices (RTI results in the appendix).

Rescaled to the probability's 0–1 range, each window's fitted Frey–Osborne slope can be compared with the published coefficient (Table 2). In the aggregate, more exposed workers support redistribution more in 2002–2014, at 0.180 (95% CI $[0.104, 0.257]$, including the published benchmark of 0.162). While the 2016–2022 aggregate is 0.020, it is not distinguishable from zero.

Next, let us consider the seventeen countries in the published sample. The aggregate follows the same pattern, at 0.145 (CI $[0.067, 0.223]$) in 2002–2014 and -0.009 in 2016–2022, when it is again indistinguishable from zero.

To estimate attenuation directly, we interact exposure with a post-2016 indicator in a single model. In the first model, the slope changes by -0.044 per standard deviation of exposure (SE 0.005) from 0.052 before 2016, and the RTI slope by -0.028 (SE 0.005). We also interact exposure with every survey year instead of a breakpoint. The second model gives $\chi^2(10) = 118.71$ for Frey–Osborne and $\chi^2(10) = 34.97$ for RTI in the joint test that all year interactions are zero.

Table 2: Global FO automation–redistribution association: published benchmark and post-estimation window aggregates of the single heterogeneity model

Quantity	Years	Scale	Estimate	95% CI	N	Countries
Published benchmark	2002–2012	native	0.162	–	59,997	17
Window aggregate of $\beta_{c,t}$	2002–2014	native	0.180	[0.104, 0.257]	115,135	31
Window aggregate of $\beta_{c,t}$	2016–2022	native	0.020	[-0.059, 0.098]	72,710	31
Full-panel aggregate of $\beta_{c,t}$	2002–2022	native	0.117	[0.042, 0.191]	187,845	32
Benchmark-roster aggregate	2002–2014	native	0.145	[0.067, 0.223]	82,081	17
Benchmark-roster aggregate	2016–2022	native	-0.009	[-0.090, 0.072]	44,085	16
Benchmark-roster aggregate	2002–2022	native	0.090	[0.014, 0.166]	126,166	17
Full-panel aggregate of $\beta_{c,t}$	2002–2022	per SD	0.039	[0.014, 0.064]	187,845	32

Window aggregates are equal-weighted means of the country-year slopes $\beta_{c,t}$ from the single multilevel heterogeneity model; no additional model is estimated. The model adjusts for relative equivalized household income, years of education, gender, age, union membership, religiosity, and unemployment status, with ESS design weights. Benchmark-roster rows restrict the aggregation to the seventeen countries of the published benchmark's sample, holding country composition fixed at the benchmark roster. Native-scale entries divide the per-SD aggregate by the pooled exposure SD. Intervals carry over the fixed-slope sampling variance and the aggregated conditional variances of the slope components; cross-component posterior covariances are ignored. The published-benchmark row is quoted from Thewissen and Rueda (2019) as an external reference.

One standard deviation of Frey–Osborne exposure is associated with 0.038 scale points of support, and the RTI coefficient is 0.027 (Table 3). In occupational terms, one standard deviation of Frey–Osborne exposure is the distance from Electrical and electronic trades workers (0.52), the occupation group at the two digit level nearest the panel mean, to Customer services clerks (0.84), the group nearest one standard deviation above it. Hence, the gradient is 1.0% of the sample mean of 3.82 on the 1–5 scale, weighted by design weights.

Skill specificity and the Walter offshorability index are the two occupational rivals of Thewissen and Rueda (2019). The exposure gradient is 8.2 times the size of the skill specificity coefficient per standard deviation under Frey–Osborne (2.3 times under RTI). Workers in offshorable occupations show lower support (-0.095), as in the benchmark study.

The gradient, compared with the economic determinants estimated beside it, is smaller, at 74% of the education effect and 32% of the relative income effect per standard deviation, but it exceeds the contrast between a temporary or absent contract and an unlimited one (0.027) and the effect of household size. Placement on the left–right scale has the strongest coefficient in the model, and the exposure gradient is 19% of it. We leave occupational unemployment risk out

of this model, since the mechanism analysis treats it as a channel from exposure to preferences and conditioning on that channel shrinks the gradient (Table 16).

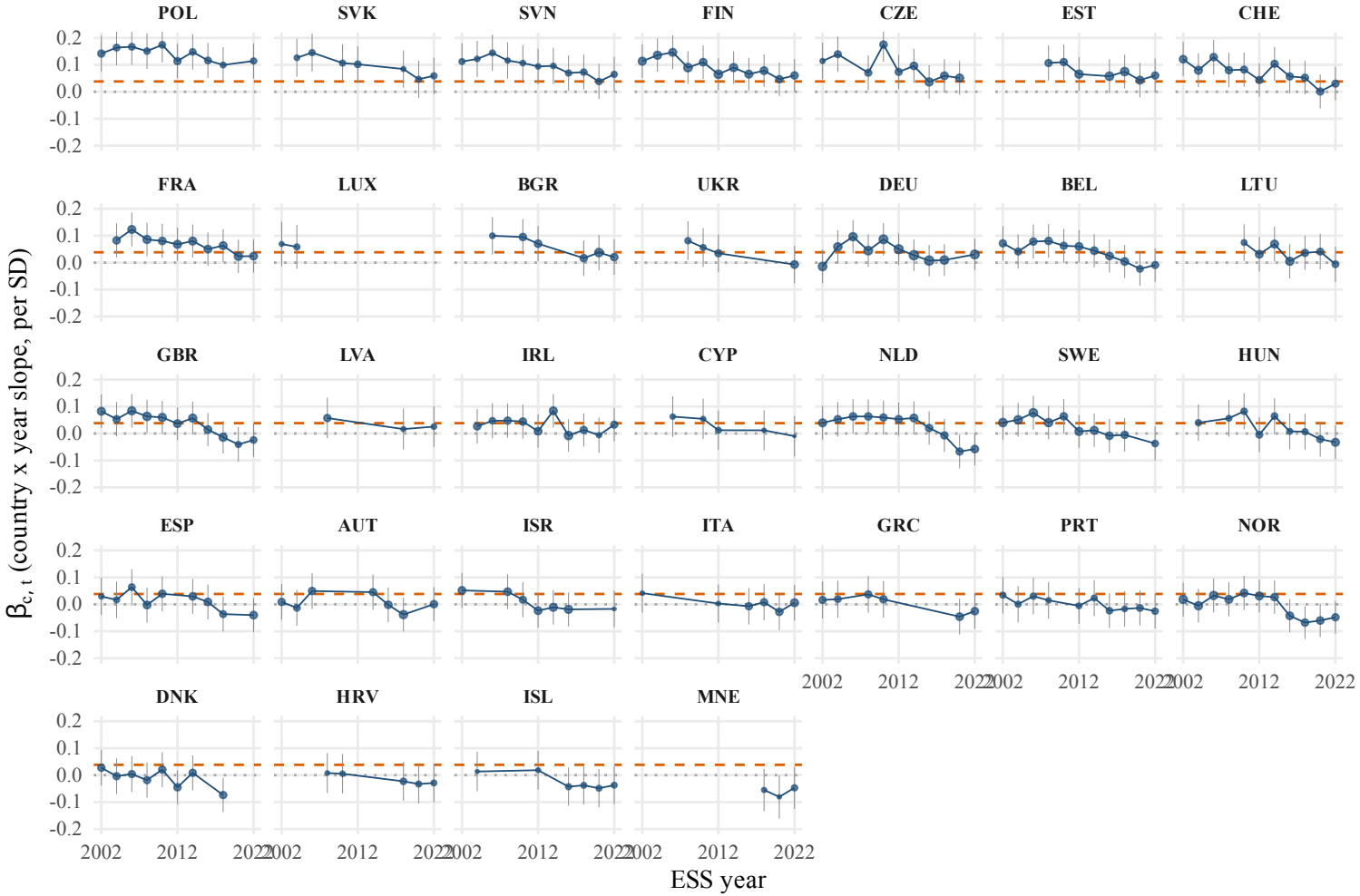
Table 3: The exposure gradient beside canonical determinants of redistribution support

	Frey–Osborne		RTI	
	Effect [95% CI]	% of mean	Effect [95% CI]	% of mean
Occupational exposure	0.038 [0.023, 0.052]	1.0	0.027 [0.017, 0.038]	0.7
Skill specificity	0.005 [-0.001, 0.010]	0.1	0.012 [0.007, 0.017]	0.3
Offshorable occupation	-0.095 [-0.105, -0.084]	2.5	-0.091 [-0.102, -0.080]	2.4
Left–right self-placement	-0.197 [-0.203, -0.192]	5.2	-0.191 [-0.196, -0.185]	5.0
Relative income	-0.117 [-0.123, -0.112]	3.1	-0.124 [-0.129, -0.118]	3.2
Education	-0.051 [-0.057, -0.045]	1.3	-0.062 [-0.068, -0.056]	1.6
Temporary or no contract	0.027 [0.013, 0.041]	0.7	0.028 [0.013, 0.042]	0.7
Household size	-0.024 [-0.029, -0.019]	0.6	-0.025 [-0.031, -0.020]	0.7

One model per exposure index: redistribution support on standardized exposure, skill specificity, an offshorable-occupation indicator, left–right self-placement, contract status, household size, and the baseline control vector, with year fixed effects, a random exposure slope by country, a country-year random intercept, and ESS design weights; occupational unemployment risk is excluded. Skill specificity and the offshorability score are the occupation-level measures of the Thewissen–Rueda replication package, mapped to both ESS occupation codings; the indicator equals one where the Walter score is at least 25, their binary convention. Continuous variables are scaled to sample standard deviations; the two category contrasts are against an unlimited contract and a non-offshorable occupation. Percentages divide the absolute effect by the design-weighted outcome mean on the 1–5 scale (3.82 FO sample, 3.81 RTI sample). $N = 138,208$ (FO), $N = 127,808$ (RTI).

The estimates for every country and year range from negative to strongly positive values (Figure 2). Variation across countries exceeds variation over time. The appendix provides the corresponding RTI plot.

Figure 2: Heterogeneity by country and year in the automation–redistribution association

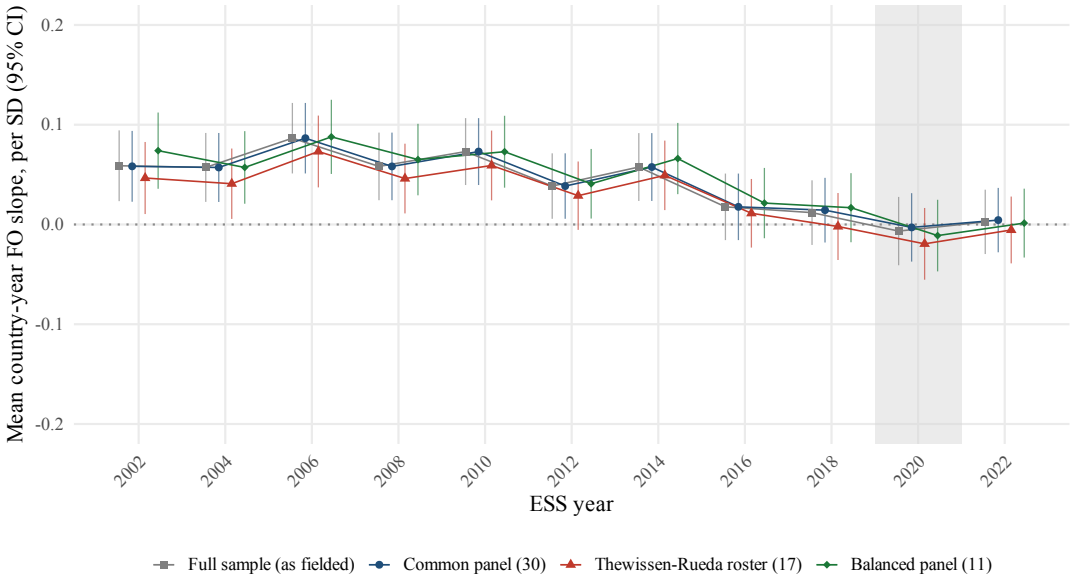


Note: Each panel is a country, ordered by mean slope; point size scales with cell N. The dashed orange line is the global slope, and the dotted line is at zero. Whiskers are 95% intervals from the fitted slopes and are cropped at the panel edge where they exceed the range. Effect axis fixed at $[-0.20, +0.20]$, shared across all slope figures.

Figure 3 separates the temporal pattern from changing country coverage. It aggregates the same fitted $\beta_{c,t}$ by year for four country rosters. The four country rosters are all observed countries, a common panel of countries present in at least three waves with a pre-2016 anchor, the seventeen countries in the Thewissen and Rueda (2019) sample, and a balanced panel of the eleven countries present in every wave. With equal country weights, differences among the curves are attributable to roster composition.

The four series differ in level, with the largest slopes in the balanced panel. However, each series declines after 2014, reaches a low point in the 2020 round, which COVID disrupted, and partly recovers in 2022. Thus, the pattern does not appear to be driven by changing country coverage. The appendix reports the RTI counterpart.

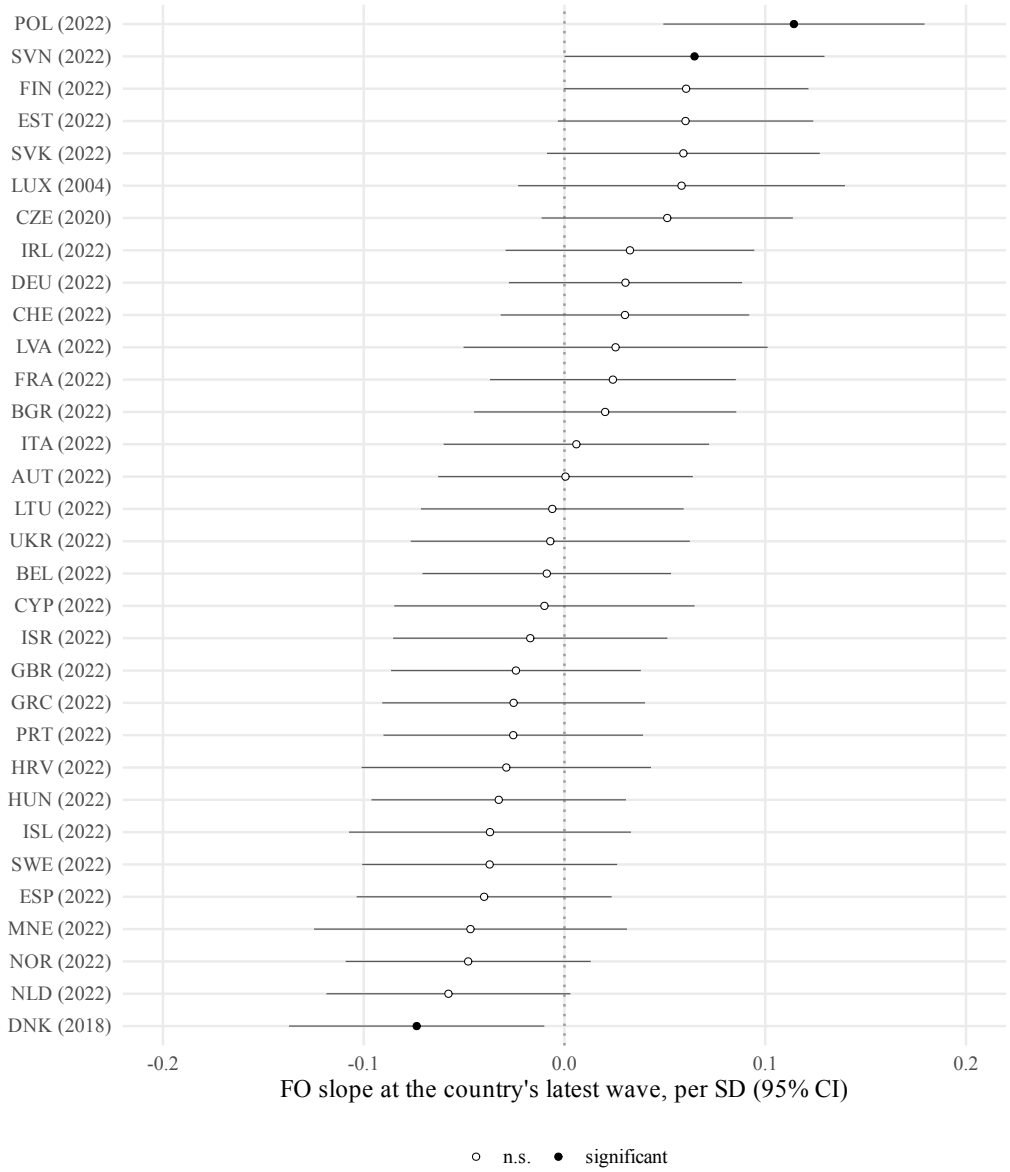
Figure 3: Yearly Frey–Osborne automation–redistribution slopes under four composition assumptions



Note: Series are offset horizontally to avoid overlap, the shaded band marks the 2020 round (ESS10), which COVID disrupted, and the dotted line is at zero. Whiskers are 95% intervals from the fitted slopes. Effect axis fixed at $[-0.20, +0.20]$, shared across all slope figures.

Most marginal effects of Frey–Osborne exposure are not statistically distinguishable from zero at each country's latest available observation (Figure 4 and the corresponding RTI figure in the appendix). The figure labels the year because 3 of the 32 countries were last observed before 2022. Significant effects are positive in Poland (0.114) and Slovenia (0.065) and negative in Denmark (-0.074). Across countries, the slopes range from -0.07 to 0.11 per standard deviation.

Figure 4: Country marginal effects of Frey–Osborne exposure at the most recent available wave

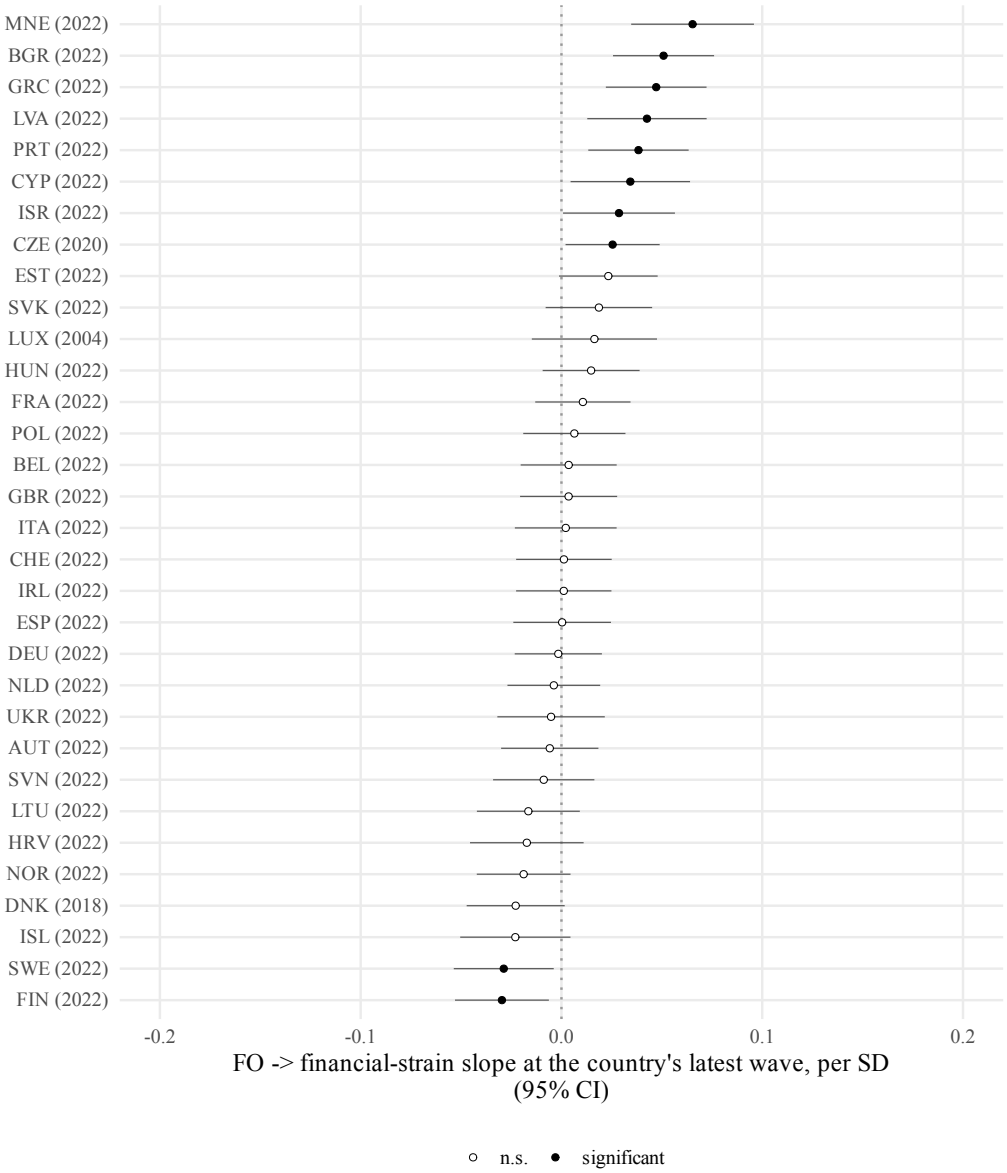


Note: Year in parentheses is each country's most recent ESS round. Whiskers are 95% intervals from the fitted slopes, and the dotted line is at zero. Effect axis fixed at $[-0.20, +0.20]$, shared across all slope figures.

With financial strain as the outcome in Figure 5, we find at the latest wave positive and significant exposure–strain slopes in eight countries, concentrated in Southern and Eastern Europe, against positive exposure–redistribution slopes in two. Portugal, Greece, Bulgaria, and Latvia show strain without demand, and Poland and Slovenia show demand without strain. The strain slope is negative in Finland (-0.030) and Sweden (-0.029). Our interpretation, under the model, is that strain without demand is consistent with uninsured exposure that still binds materially where demand has faded, whereas demand without strain is consistent with institu-

tions that absorb the material consequence while political demand persists.

Figure 5: Country marginal effects of Frey–Osborne exposure on financial strain at the most recent available wave



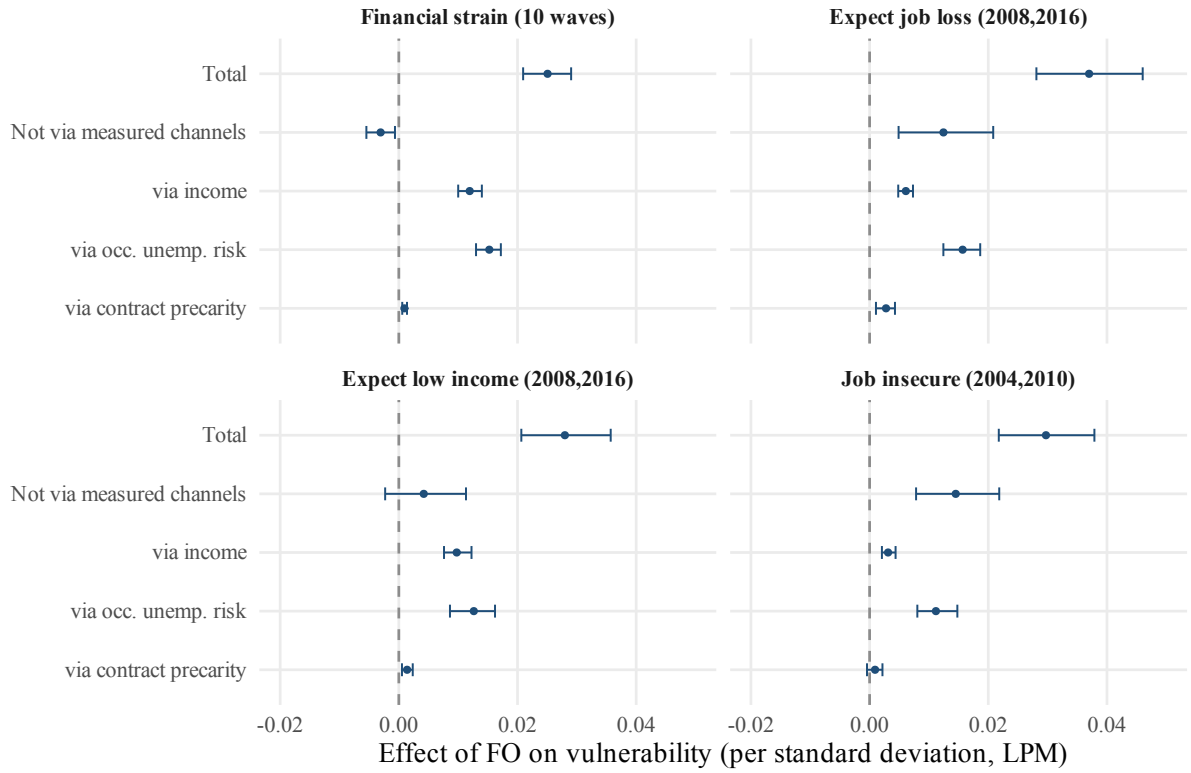
Note: Year in parentheses is each country's most recent ESS round. Whiskers are 95% intervals from the fitted strain slopes, and the dotted line is at zero. Effect axis fixed at $[-0.20, +0.20]$, shared across all slope figures.

5 Mechanism Analyses: Vulnerability and Institutional Moderation

The paper's main estimand is the association between automation exposure and redistribution preferences. Figure 6 presents the mechanism results at the individual level. Automation expo-

sure predicts financial strain across the 10 waves in which every channel is measured. Exposure also predicts expected job loss, expected income insufficiency, and job insecurity in the rounds that include them. Appendix Table 6 reports the corresponding estimates for the 2008 and 2016 rounds with the welfare module, in which the 2016 columns are the confirmatory specification and the 2008 columns are an extension under the same model. The strain gradient and the redistribution gradient attenuate over the same waves. This is consistent with a common political cause and equally consistent with decay in the exposure measure itself, since a measure that tracks occupational risk less well over time would weaken both.

Figure 6: Automation exposure and perceived economic vulnerability across ESS waves



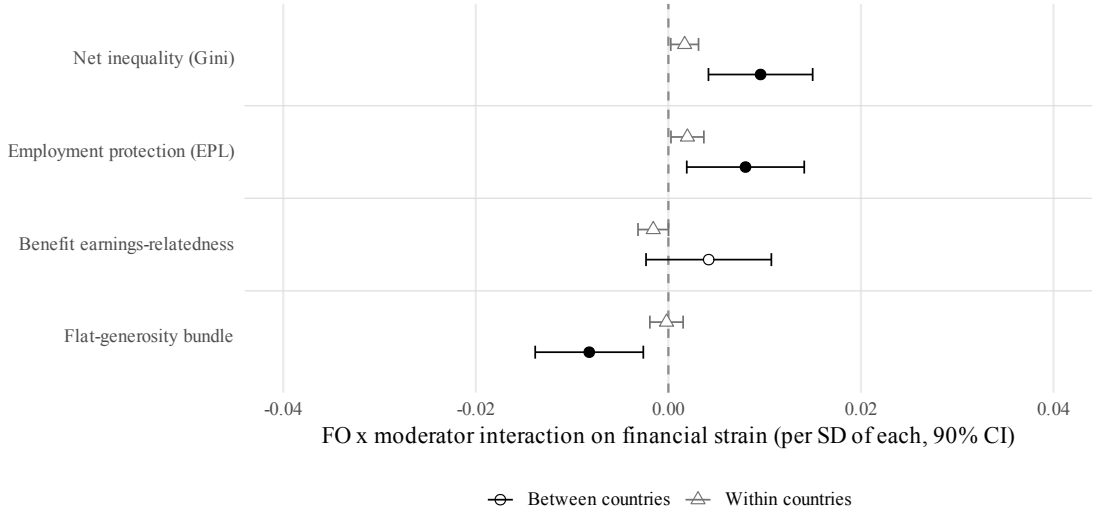
Note: Facets are the vulnerability outcomes. Within each, rows decompose the association into a direct part (not via measured channels) and indirect parts through each channel (total = direct + channels). Whiskers are 95% intervals from a cluster bootstrap. Effect axis fixed at $[-0.02, +0.05]$, shared across mediation figures.

Figure 7 examines whether institutions condition the association between automation risk and financial strain, displaying the components of each moderation between and within countries separately. The figure reports the four moderators that Table 1 maps to the theory.

Benefits and spending expand in downturns, when the strain gradient also steepens. Without accounting for the business cycle, we would therefore miss a buffering pattern. After adjust-

ment for the cycle, the results for the focal moderators indicate a flatter gradient in countries with a larger flat generosity bundle, at -0.008 ($q = 0.064$), and a steeper one in countries with stricter employment protection, at 0.008 ($q = 0.082$), and higher net inequality, at 0.010 ($q = 0.029$). Each equals between 52% and 63% of the average gradient per standard deviation of the moderator. These moderations are large in relative terms because the strain gradient itself is small, at 0.015 per standard deviation of exposure on average across these models. Under Frey–Osborne, the components within countries fail to reach significance for strain after adjustment. Under RTI, benefit earnings relatedness has a significant component within countries (-0.002, $q = 0.068$). Estimated with the two stage diagnostic, the pooled models with a single interaction, and the full sweep, the moderations retain these signs (see appendix). Consistent with insider–outsider dualization, the gradient steepens under strict employment protection. In that account, strict protection shelters insiders and leaves exposed outsiders more precarious.

Figure 7: Institutional moderation of the association between automation and financial strain

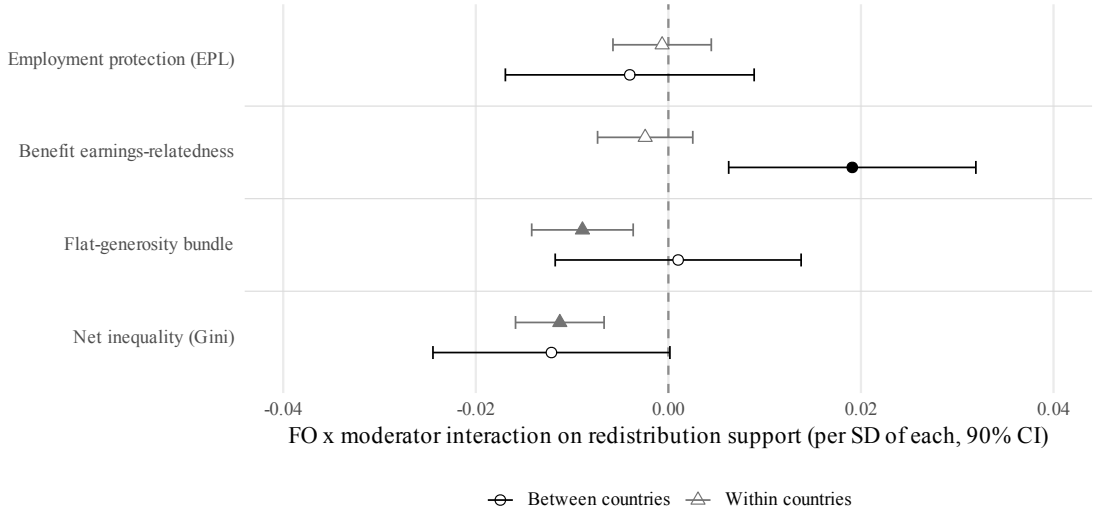


Note: Circles mark the component between countries (country mean of the lagged moderator), and triangles the component within countries (deviation from it). Filled symbols mark Benjamini–Hochberg $q < 0.10$, and whiskers are unadjusted 90% CIs. Effect axis fixed at $[-0.04, +0.04]$, shared across interaction figures.

As net inequality rises within a country, the redistribution gradient becomes flatter (-0.011, $q = 0.001$, Figure 8), by 31% of the average gradient (0.037) per standard deviation of the moderator. Benefit earnings relatedness amplifies it between countries ($+0.019$, $q = 0.077$, 52% of the gradient), the sign the retention share requires. The flat generosity bundle buffers it within countries (-0.009, $q = 0.043$). Employment protection shows no effect. Its components sit near

zero under Frey–Osborne, and under RTI the negative interaction within countries (-0.007) does not reach significance after adjustment ($q = 0.129$). Hence, the four focal tests support all three signed predictions, and the null for employment protection is what Proposition 4 predicts at the risk aversion the income results adopt. While the components within countries are the stronger evidence because country confounds that do not change over time drop out of them, a coefficient within countries explains the temporal attenuation only if the moderator itself also moved.

Figure 8: Institutional moderation of the exposure–redistribution gradient between and within countries



Note: Circles mark the component between countries (country mean of the lagged moderator), and triangles the component within countries (deviation from it).

Appendix 8.2 gives additional mechanism diagnostics, robustness checks, and full auxiliary tables.

6 Discussion

The findings qualify a simple account, constant over time, of the political consequences of technological change. The association between automation risk and redistribution preferences replicates prior research in the pre-2016 period, and, in the aggregate, it all but disappears in the subsequent decade. Changing country coverage and sample composition do not account for the decline, so something changed over time in most of the countries where the association ini-

tially held, while a few countries never showed it at all. The switch in occupation coding cannot account for this timing either. The ESS moved from ISCO-88 to ISCO-08 in 2012. Yet, under the new coding, the association remains positive and significant in both 2012 and 2014, with the 2014 slope matching the levels seen under the old regime (Appendix Figure 12). If the coding switch alone drove the decline, we would expect null estimates from 2012 onward, but they only begin in 2016, two rounds after the switch and within an unchanged coding regime.

Individual countries illustrate the pattern (Figure 2). France shows significant positive slopes in seven of eight rounds from 2004 through 2018 and null estimates in 2020 and 2022. In Great Britain, we find the same fading association earlier, with significant positive slopes through 2008, null estimates from 2010 onward, and negative point estimates after 2016. Finland has a significant positive slope in every round from 2002 through 2018. Spain never has a significant slope in any round. Differences in insurance parameters can produce these paths in the model. For example, we would expect a flatter demand gradient with a shift toward flat and conditional transfers, as in the welfare reforms of the United Kingdom during the 2010s. Encompassing systems tied to earnings, like Finland's, would preserve the gradient. Thus, we find divergence consistent with these mechanisms, although our design does not allow strict causal attribution of any single country's path.

A shift in the nature of technological risk itself may link this variation across countries and the aggregate decline over time. Technological change biased toward routine work in the 2000s was concentrated among clerical and manual occupations, whereas generative AI now exposes professionals with higher skills and incomes to technological precarity (Gallego and Kurer, 2022). Note that we measure only part of this shift. For the benchmark roster, Appendix Figure 15 shows a steady decline across rounds in routine exposure when we use design weights. Since Frey–Osborne and RTI no longer measure where the binding economic risk lies, these canonical indices may no longer reliably predict general left wing political preferences. Professionals with higher incomes also face different incentives once they become exposed, incentives that change the institutional mechanics of redistribution demand. These workers have weaker material incentives for conventional income transfers and favor instruments such as social investment or basic income instead (Busemeyer and Sahm, 2022).

We find an association between workers' automation risk and their perceived vulnerability, including financial strain and expected job loss. Through these mechanism analyses, we can better understand where the association with redistribution concentrates.

After accounting for macroeconomic cycles, we find patterns consistent with the social insurance account of redistribution demand (Moene and Wallerstein, 2001). Across countries, generous flat transfers predict a flatter association between exposure and strain, while high net income inequality predicts a steeper one. Under Frey–Osborne, the corresponding estimates for strain within countries are small and imprecise. We note that the institutional associations for redistribution appear both between and within countries, a contrast between the two outcomes that we do not attempt to resolve.

Our estimates rely on indices at the occupation level (Frey–Osborne probabilities and RTI), under the assumption of uniform task structures within broad occupations. However, harmonizing ISCO-88 and ISCO-08 introduces unavoidable crosswalk measurement error. Although changing country coverage across ESS waves prevents us from using a perfectly balanced panel, we address this empirical limitation with models on a common sample that adjust for composition. We also do not draw strict causal conclusions, since our design is observational.

7 Conclusion

The main objective of this paper was to reexamine the relationship between occupational automation risk and demand for redistribution in harmonized European Social Survey data for 2002–2022. We find that the widely cited historical association between automation exposure and redistribution preferences replicates before 2016 and all but disappears from 2016 onward. This attenuation differs far more across countries than over time. Thus, the collapse of this political coalition comes from divergent national trajectories, not from a uniform continental shock.

Our formal insurance model grounds these empirical patterns in structural institutional differences rather than ad hoc psychological shifts. Hence, changing institutional parameters rather than a break in political economic logic account for the attenuation over time and the divergence across countries.

Future research must use new exposure measures to determine whether highly skilled professionals exposed to generative AI now face the same political economic logic that previously governed routine workers. To map this new terrain, future surveys must measure the reentry wage after displacement and the expected duration of unemployment, the components of institutional buffering that our model identifies, rather than rely on broad occupational aggregates. Finally, tracking these shifting political coalitions will require studying institutional reforms within countries over time, not only comparisons of countries at one point in time.

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Online Appendix

Automation Risk, Redistribution, and Economic Vulnerability

The Online Appendix, Section 8 of this paper, contains the formal insurance model of redistribution preferences and the supplementary figures and tables. It is available at

www.ignacio-urbina.com/files/automation-redistribution-appendix.pdf